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Gender Affirming Surgery

Number: 0615

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Scope of Policy

This Clinical Policy Bulletin addresses gender affirming surgery.

Note: Some plans may cover gender affirming procedures in addition to the following policy. Alternatively, some plans may not cover gender affirming procedures in the following policy. Please check the specific benefit plan documents.

I. **Medical Necessity**

Aetna considers gender affirming surgery medically necessary when criteria for each of the following procedures are met:

A. **Requirements for Breast Removal**

1. Signed letter from a qualified mental health professional (see Appendix) assessing the transgender/gender diverse individual's readiness for physical treatment; *and*
2. Documentation of marked and sustained gender dysphoria (see Appendix); *and*
3. Other possible causes of apparent gender incongruence have been excluded; *and*
4. Mental and physical health conditions that could negatively impact the outcome of gender-affirming medical treatments are assessed, with risks and benefits discussed; *and*
5. Capacity to consent for the specific physical treatment; *and*
6. For members less than 18 years of age, completion of one year of testosterone treatment, unless hormone therapy is not desired or medically contraindicated; *and*
7. Risk factors associated with breast cancer have been assessed.

B. **Requirements for Breast Augmentation (Implants/Lipofilling)**

1. Signed letter from a qualified mental health professional (see Appendix) assessing the transgender/gender diverse individual's readiness for physical treatments; *and*
2. Documentation of marked and sustained gender dysphoria (see Appendix); *and*
3. Other possible causes of apparent gender incongruence have been excluded; *and*

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4. Mental and physical health conditions that could negatively impact the outcome of gender-affirming medical treatments are assessed, with risks and benefits discussed; *and*
5. Capacity to consent for the specific physical treatment; *and*
6. Completion of six months of feminizing hormone therapy (12 months for adolescents less than 18 years of age) prior to breast augmentation surgery, unless hormone therapy is not desired or medically contraindicated); *and*
7. Risk factors associated with breast cancer have been assessed.

C. **Requirements for Gonadectomy (Hysterectomy and Oophorectomy or Orchiectomy)**

1. Signed letter from a qualified mental health professional (see Appendix) assessing the transgender/gender diverse individual's readiness for physical treatments; *and*
2. Documentation of marked and sustained gender dysphoria (see Appendix); *and*
3. Other possible causes of apparent gender incongruence have been excluded; *and*
4. Mental and physical health conditions that could negatively impact the outcome of gender-affirming medical treatments are assessed, with risks and benefits discussed; *and*
5. Capacity to consent for the specific physical treatment; *and*
6. Six months of continuous hormone therapy as appropriate to the member's gender goals (12 months for adolescents less than 18 years of age), unless hormone therapy is not desired or medically contraindicated.

D. **Requirements for Genital Reconstructive Surgery (i.e., vaginectomy, urethroplasty, metoidioplasty, phalloplasty, scrotoplasty, placement of a testicular prosthesis and erectile prosthesis, penectomy, vaginoplasty, labiaplasty, clitoroplasty and electrolysis or laser hair removal sessions for skin graft preparation for genital surgery)**

1. Signed letter from a qualified mental health professional (see Appendix) assessing the transgender/gender diverse individual's readiness for physical treatments; *and*
2. Documentation of marked and sustained gender dysphoria (see Appendix); *and*
3. Other possible causes of apparent gender incongruence have been excluded; *and*
4. Mental and physical health conditions that could negatively impact the outcome of gender-affirming medical treatments are assessed, with risks and benefits discussed; *and*
5. Capacity to consent for the specific physical treatment; *and*
6. Six months of continuous hormone therapy as appropriate to the member's gender goals (12 months for adolescents less than 18 years of age), unless hormone therapy is not desired or medically contraindicated.

Note on gender specific services for the transgender community: Gender-specific services may be medically necessary for transgender persons appropriate to their anatomy. Examples include:

1. Breast cancer screening may be medically necessary for transmasculine persons who have not undergone chest masculinization surgery;
2. Prostate cancer screening may be medically necessary for transfeminine persons who have retained their prostate.

Aetna considers reversal of gender affirming surgery (performing surgical procedures to return anatomy to that of the sex assigned at birth) medically necessary for persons who regret their gender-related surgical intervention, where applicable requirements for gender affirming surgery listed above are met.

Aetna considers gonadotropin-releasing hormone medically necessary to suppress puberty in trans identified adolescents if they meet World Professional Association for Transgender Health (WPATH) criteria (see CPB 0501 - Gonadotropin-Releasing Hormone Analogs and Antagonists).

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Aetna considers gender affirming breast removal revision surgery medically necessary for persons who meet the criteria for gender affirming breast removal surgery and have documentation, including high-quality color photographs, that support the medical necessity for the intended revision surgery.

II. Not Medically Necessary

Aetna considers more than one breast augmentation not medically necessary. This does not include the medically necessary replacement of breast implants (see CPB 0142 - Breast Implant Removal).

Aetna considers nerve grafting for gender affirming breast surgery not medically necessary.

Aetna considers the following procedures that may be performed as a component of a gender transition as not medically necessary and cosmetic (not an all-inclusive list) (see also CPB 0031 - Cosmetic Surgery):

- Hair removal (e.g., electrolysis, laser hair removal) (Exception: A limited number of electrolysis or laser hair removal sessions are considered medically necessary for skin graft preparation for genital surgery)
- Tracheal shave (reduction thyroid chondroplasty)
- Facial Gender Affirming Procedures, including:
    - Brow (reduction, augmentation, lift)
    - Hair line advancement and/or hair transplant
    - Facelift/mid-face lift (following alteration of the underlying skeletal structures) (platysmaplasty)
    - Blepharoplasty (lipofilling)
    - Rhinoplasty (+/- fillers)
    - Cheek (implant, lipofilling)
    - Lip (upper lip shortening, lip augmentation)
    - Lower jaw (reduction of mandibular angle, augmentation)
    - Chin reshaping (osteoplastic, alloplastic (implant-based))
    - Chondrolaryngoplasty (also known as Adam's apple reduction, thyroid cartilage reduction, or tracheal shave)
    - Vocal cord surgery
- Body contouring gender affirming surgery, including (not an all-inclusive list):
    - Liposuction/lipofilling/implants (pectoral, hip, gluteal, calf).

III. Experimental, Investigational, or Unproven

Aetna considers the following experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

- Use of facial recognition neural networks for facial feminization surgery;
- Frontal contouring with corrugator resection for facial feminization surgery.

IV. Related Policies

- CPB 0031 - Cosmetic Surgery
- CPB 0097 - External Breast Prosthesis
- CPB 0501 - Gonadotropin-Releasing Hormone Analogs and Antagonists
- CPB 0646 - Voice Therapy

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Table:

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered if selection criteria are met:	
13131	Repair, complex, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands and/or feet; 1.1 cm to 2.5 cm
13132	Repair, complex, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands and/or feet; 2.6 cm to 7.5 cm
13133	Repair, complex, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands and/or feet; each additional 5 cm or less
13160	Secondary closure of surgical wound or dehiscence, extensive or complicated
14021	Adjacent tissue transfer or rearrangement, scalp, arms and/or legs; defect 10.1 sq cm to 30.0 sq cm
14040	Adjacent tissue transfer or rearrangement, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands and/or feet; defect 10 sq cm or less
14041	Adjacent tissue transfer or rearrangement, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands and/or feet; defect 10.1 sq cm to 30.0 sq cm
14301	Adjacent tissue transfer or rearrangement, any area; defect 30.1 sq cm to 60.0 sq cm
14302	Adjacent tissue transfer or rearrangement, any area; each additional 30.0 sq cm, or part thereof
15002 -15003	Surgical preparation or creation of recipient site by excision of open wounds, burn eschar, or scar (including subcutaneous tissues), or incisional release of scar contracture, trunk, arms, legs; first 100 sq cm or 1% of body area of infants and children. + each additional
15004	Surgical preparation or creation of recipient site by excision of open wounds, burn eschar, or scar (including subcutaneous tissues), or incisional release of scar contracture, face, scalp, eyelids, mouth, neck, ears, orbits, genitalia, hands, feet and/or multiple digits; first 100 sq cm or 1% of body area of infants and children
15100 - 15101	Split-thickness autograft, trunk, arms, legs; first 100 sq cm or less, or 1% of body area of infants and children + each additional 1%
15115	Epidermal autograft, face, scalp, eyelids, mouth, neck, ears, orbits, genitalia, hands, feet, and/or multiple digits; first 100 sq cm or less, or 1% of body area of infants and children
15120	Split-thickness autograft, face, scalp, eyelids, mouth, neck, ears, orbits, genitalia, hands, feet, and/or multiple digits; first 100 sq cm or less, or 1% of body area of infants and children
15240 - 15241	Full thickness graft, free, including direct closure of donor site, forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands, and/or feet; 20 sq cm or less. + each additional
15273 -15274	Application of skin substitute graft to trunk, arms, legs, total wound surface area greater than or equal to 100 sq cm; first 100 sq cm wound surface area, or 1% of body area of infants and children + each additional 1%
15275	Application of skin substitute graft to face, scalp, eyelids, mouth, neck, ears, orbits, genitalia, hands, feet, and/or multiple digits, total wound surface area up to 100 sq cm; first 25 sq cm or less wound surface area

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Code	Code Description
15277 - 15278	Application of skin substitute graft to face, scalp, eyelids, mouth, neck, ears, orbits, genitalia, hands, feet, and/or multiple digits, total wound surface area greater than or equal to 100 sq cm; first 100 sq cm wound surface area, or 1% of body area of infants and children. + each additional 1%
15574	Formation of direct or tubed pedicle, with or without transfer; forehead, cheeks, chin, mouth, neck, axillae, genitalia, hands or feet
15734	Muscle, myocutaneous, or fasciocutaneous flap; trunk
15738	Muscle, myocutaneous, or fasciocutaneous flap; lower extremity
15740	Flap; island pedicle requiring identification and dissection of an anatomically named axial vessel
15750	Flap; neurovascular pedicle
15757	Free skin flap with microvascular anastomosis
15771	Grafting of autologous fat harvested by liposuction technique to trunk, breasts, scalp, arms, and/or legs; 50 cc or less injectate [covered for breast augmentation only]
15772	Grafting of autologous fat harvested by liposuction technique to trunk, breasts, scalp, arms, and/or legs; each additional 50 cc injectate, or part thereof (List separately in addition to code for primary procedure) [covered for breast augmentation only]
15773	Grafting of autologous fat harvested by liposuction technique to face, eyelids, mouth, neck, ears, orbits, genitalia, hands, and/or feet; 25 cc or less injectate
15860	Intravenous injection of agent (eg, fluorescein) to test vascular flow in flap or graft
17380	Electrolysis epilation, each 30 minutes [Check benefits]
17999	Unlisted procedure, skin, mucous membrane and subcutaneous tissue [laser hair removal] [Check benefits]
19318	Reduction mammaplasty
19325	Breast augmentation with implant
19350	Nipple/areola reconstruction [only covered when not performed at time of original breast surgery]
19357	Tissue expander placement in breast reconstruction, including sub sequent expansion(s) can be authorized for gender affirmation coverage
19380	Revision of reconstructed breast (eg, significant removal of tissue, re-advancement and/or re-inset of flaps in autologous reconstruction or significant capsular revision combined with soft tissue excision in implant-based reconstruction)
40808	Biopsy, vestibule of mouth
40818	Excision of mucosa of vestibule of mouth as donor graft
49329	Unlisted laparoscopy procedure, abdomen, peritoneum and omentum [graft from colon for vaginoplasty]
51040	Cystostomy, cystotomy with drainage
51102	Aspiration of bladder; with insertion of suprapubic catheter
52005	Cystourethroscopy, with ureteral catheterization, with or without irrigation, instillation, or ureteropyelography, exclusive of radiologic service
53400	Urethroplasty; first stage, for fistula, diverticulum, or stricture (eg, Johannsen type)
53405	Urethroplasty; second stage (formation of urethra), including urinary diversion
53410	Urethroplasty, 1-stage reconstruction of male anterior urethra
53430	Urethroplasty, reconstruction of female urethra
53520	Closure of urethrostomy or urethrocutaneous fistula, male (separate procedure)
54120	Amputation of penis; partial
54125	Amputation of penis; complete
54235	Injection of corpora cavernosa with pharmacologic agent(s) (eg, papaverine, phentolamine)

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Code	Code Description
54300	Plastic operation of penis for straightening of chordee (eg, hypospadias), with or without mobilization of urethra
54304	Plastic operation on penis for correction of chordee or for first stage hypospadias repair with or without transplantation of prepuce and/or skin flaps
54336	1-stage perineal hypospadias repair requiring extensive dissection to correct chordee and urethroplasty by use of skin graft tube and/or island flap
54400 - 54417	Penile prosthesis
54520	Orchiectomy, simple (including subcapsular), with or without testicular prosthesis, scrotal or inguinal approach
54660	Insertion of testicular prosthesis (separate procedure)
55150	Resection of scrotum
55175	Scrotoplasty; simple
55180	complicated
55970	Intersex surgery; male to female [a series of staged procedures that includes male genitalia removal, penile dissection, urethral transposition, creation of vagina and labia with stent placement]
55980	female to male [a series of staged procedures that include penis and scrotum formation by graft, and prostheses placement]
56625	Vulvectomy simple; complete
56800	Plastic repair of introitus
56805	Clitoroplasty for intersex state
56810	Perineoplasty, repair of perineum, nonobstetrical (separate procedure)
57106, 57110	Vaginectomy, partial removal of vaginal wall, or complete removal of vaginal wall
57282	Colpopexy, vaginal; extra-peritoneal approach (sacrospinous, iliococcygeus)
57291 - 57292	Construction of artificial vagina
57335	Vaginoplasty for intersex state
57425	Laparoscopy, surgical, colpopexy (suspension of vaginal apex)
58150, 58180, 58260 - 58262, 58275 - 58291, 58541 - 58544, 58550 - 58554	Hysterectomy
58570 - 58573	Laparoscopy, surgical, with total hysterectomy
58661	Laparoscopy, surgical; with removal of adnexal structures (partial or total oophorectomy and/or salpingectomy)
58720	Salpingo-oophorectomy, complete or partial, unilateral or bilateral
58999	Unlisted procedure, female genital system (nonobstetrical) [metoidioplasty]
64708	Neuroplasty, major peripheral nerve, arm or leg, open; other than specified
64856	Suture of major peripheral nerve, arm or leg, except sciatic; including transposition
64859	Suture of each additional major peripheral nerve
64874	Suture of nerve; requiring extensive mobilization, or transposition of nerve
64910	Nerve repair; with synthetic conduit or vein allograft (eg, nerve tube), each nerve

CPT codes not covered for indications listed in the CPB [considered not medically necessary and cosmetic]:

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Code	Code Description
***Facial recognition neural networks* -no specific code**	
11950 - 11954	Subcutaneous injection of filling material (e.g., collagen)
15200	Full thickness graft, free, including direct closure of donor site, trunk; 20 sq cm or less [nipple reconstruction]
15775	Punch graft for hair transplant; 1 to 15 punch grafts
15776	Punch graft for hair transplant; more than 15 punch grafts
15780 - 15787	Dermabrasion
15788 - 15793	Chemical peel
15820 - 15823	Blepharoplasty
15824 - 15828	Rhytidectomy [face-lifting]
15830 - 15839	Excision, excessive skin and subcutaneous tissue (includes lipectomy); abdomen, infraumbilical panniculectomy
15876 - 15879	Suction assisted lipectomy
17380	Electrolysis epilation, each 30 minutes
19301	Mastectomy, partial (eg, lumpectomy, tylectomy, quadrantectomy, segmentectomy)
19303	Mastectomy, simple, complete
19316	Mastopexy
19340	Immediate insertion of breast prosthesis following mastopexy, mastectomy or in reconstruction
19342	Delayed insertion of breast prosthesis following mastopexy, mastectomy or in reconstruction
20999	Unlisted procedure, musculoskeletal system, general [unlisted augmentation] [check benefits]
21087	Nasal prosthesis
21120 - 21123	Genioplasty
21125 - 21127	Augmentation, mandibular body or angle; prosthetic material or with bone graft, onlay or interpositional (includes obtaining autograft)
21137	Reduction forehead; contouring only
21138	Reduction forehead; contouring and application of prosthetic material or bone graft (includes obtaining autograft)
21139	Reduction forehead; contouring and setback of anterior frontal sinus wall
21193	Reconstruction of mandibular rami, horizontal, vertical, C, or L osteotomy; without bone graft
21194	with bone graft (includes obtaining graft)
21195	Reconstruction of mandibular rami and/or body, sagittal split; without internal rigid fixation
21196	with internal rigid fixation
21208	Osteoplasty, facial bones; augmentation (autograft, allograft, or prosthetic implant)
21210	Graft, bone; nasal, maxillary or malar areas (includes obtaining graft)
21270	Malar augmentation, prosthetic material
30400 - 30420	Rhinoplasty; primary
30430 - 30450	Rhinoplasty; secondary
31599	Unlisted procedure, larynx [thyroid chondroplasty and tracheal shave] [voice modification surgery] [check benefits]
31899	Unlisted procedure, trachea, bronchi [thyroid chondroplasty and tracheal shave] [augmentation thyroid chondroplasty (thyroid cartilage augmentation)] [check benefits][Chondrolaryngoplasty, Adam's apple reduction, thyroid cartilage reduction]
40799	Unlisted procedure, lips [lip shortening] [check benefits]

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Code	Code Description
64911	Nerve repair; with autogenous vein graft (includes harvest of vein graft), each nerve
64912	with nerve allograft, each nerve, first strand (cable)
64913	with nerve allograft, each additional strand (List separately in addition to code for primary procedure)
67900	Repair of brow ptosis (supraciliary, mid-forehead or coronal approach)
92507	Treatment of speech, language, voice, communication, and/or auditory processing disorder; individual
92508	Treatment of speech, language, voice, communication, and/or auditory processing disorder; group, two or more individuals

Other CPT codes related to the CPB:

11980	Subcutaneous hormone pellet implantation (implantation of estradiol and/or testosterone pellets beneath the skin)
+90785	Interactive complexity (List separately in addition to the code for primary procedure)
90832 - 90838	Psychotherapy
96372	Therapeutic, prophylactic, or diagnostic injection (specify substance of drug); subcutaneous or intramuscular

HCPCS codes covered if selection criteria are met:

A4280	Adhesive skin support attachment for use with external breast prosthesis, each
C1789	Prosthesis, breast (implantable)
C1813	Prosthesis, penile, inflatable
C2622	Prosthesis, penile, non-inflatable
J0172	Injection, testosterone cypionate (azmiro), 1 mg
J0173	Testosterone pellet, implant, 75 mg
J1000	Injection, depo-estradiol cypionate, up to 5 mg
J1050	Injection, medroxyprogesterone acetate, 1 mg
J1071	Injection, testosterone cypionate, 1 mg
J0172	Injection, testosterone cypionate (azmiro), 1 mg
J1380	Injection, estradiol valerate, up to 10 mg
J1726	Injection, hydroxyprogesterone caproate, (makena), 10 mg
J1729	Injection, hydroxyprogesterone caproate, not otherwise specified, 10 mg
J1950	Injection, leuprolide acetate (for depot suspension), per 3.75 mg
J1951	Injection, leuprolide acetate for depot suspension (fensolvi), 0.25 mg
J1952	Leuprolide injectable, camcevi, 1 mg
J1954	Injection, leuprolide acetate for depot suspension (cipla), 7.5 mg
J2675	Injection, progesterone, per 50 mg
J3121	Injection, testosterone enanthate, 1 mg
J3145	Injection, testosterone undecanoate, 1 mg
J3315	Injection, triptorelin pamoate, 3.75 mg
J7179	Injection, von willebrand factor (recombinant), (vonvendi), 1 i.u. vwf:rco
J7294	Segesterone acetate and ethinyl estradiol 0.15mg, 0.013mg per 24 hours; yearly vaginal system, each
J7295	Ethinyl estradiol and etonogestrel 0.015mg, 0.12mg per 24 hours; monthly vaginal ring, each
J9202	Goserelin acetate implant, per 3.6 mg

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Code	Code Description
J9217	Leuprolide acetate (for depot suspension), 7.5 mg
J9218	Leuprolide acetate, per 1 mg
J9219	Leuprolide acetate implant, 65 mg
J9226	Histrelin implant (supprelin la), 50 mg
L8000	Breast prosthesis, mastectomy bra, without integrated breast prosthesis form, any size, any type
L8001	Breast prosthesis, mastectomy bra, with integrated breast prosthesis form, unilateral, any size, any type
L8002	Breast prosthesis, mastectomy bra, with integrated breast prosthesis form, bilateral, any size, any type
L8010	Breast prosthesis, mastectomy sleeve
L8015	External breast prosthesis garment, with mastectomy form, post mastectomy
L8020	Breast prosthesis, mastectomy form
L8030	Breast prosthesis, silicone or equal, without integral adhesive
L8031	Breast prosthesis, silicone or equal, with integral adhesive
L8035	Custom breast prosthesis, post mastectomy, molded to patient model
L8039	Breast prosthesis, not otherwise specified
L8600	Implantable breast prosthesis, silicone or equal
S0138	Finasteride, 5 mg
S2066	Breast reconstruction with gluteal artery perforator (gap) flap, including harvesting of the flap, microvascular transfer, closure of donor site and shaping the flap into a breast, unilateral
S2067	Breast reconstruction of a single breast with "stacked" deep inferior epigastric perforator (diep) flap(s) and/or gluteal artery perforator (gap) flap(s), including harvesting of the flap(s), microvascular transfer, closure of donor site(s) and shaping the flap into a breast, unilateral
S2068	Breast reconstruction with deep inferior epigastric perforator (diep) flap or superficial inferior epigastric artery (siea) flap, including harvesting of the flap, microvascular transfer, closure of donor site and shaping the flap into a breast, unilateral
S4988	Penile contracture device, manual, greater than 3 lbs traction force
S8080	Scintimammography (radioimmunoscintigraphy of the breast), unilateral, including supply of radiopharmaceutical
S9560	Home injectable therapy; hormonal therapy (e.g.; leuprolide, goserelin), including administrative services, professional pharmacy services, care coordination, and all necessary supplies and equipment (drugs and nursing visits coded separately), per diem
T4545	Incontinence product, disposable, penile wrap, each

HCPCS codes not covered for indications listed in the CPB:

G0153	Services performed by a qualified speech-language pathologist in the home health or hospice setting, each 15 minutes
L8499	Unlisted procedure for miscellaneous prosthetic services [prosthetic implant] [check benefits]
L8699	Prosthetic implant, not otherwise specified [check benefits]
S9128	Speech therapy, in the home, per diem

ICD-10 codes covered if selection criteria are met:

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Code	Code Description
F64.0 - F64.9	Gender identity disorders
Z87.890	Personal history of sex reassignment

Background

The International Classification of Diseases and Related Health Problems, Tenth Revision (ICD-10) and the Diagnostic and Statistical Manual of Mental Disorders, (DSM-5-TR) are the diagnostic classifications and criteria manuals used in the United States. Notwithstanding, the World Professional Association of Transgender Health Standard of Care 8th edition (WPATH SOC8) states: "While Gender Dysphoria (GD) is still considered a mental health condition in the Diagnostic and Statistical Manual of Mental Disorders, (DSM-5-TR) of the American Psychiatric Association. Gender incongruence is no longer seen as pathological or a mental disorder in the world health community. Gender Incongruence is recognized as a condition in the International Classification of Diseases and Related Health Problems, 11th Version of the World Health Organization (ICD-11). Because of historical and current stigma, TGD people can experience distress or dysphoria that may be addressed with various gender-affirming treatment options. While nomenclature is subject to change and new terminology and classifications may be adopted by various health organizations or administrative bodies, the medical necessity of treatment and care is clearly recognized for the many people who experience dissonance between their sex assigned at birth and their gender identity."

Gender dysphoria refers to discomfort or distress that is caused by a discrepancy between an individual's gender identity and the gender assigned at birth (and the associated gender role and/or primary and secondary sex characteristics). A diagnosis of gender dysphoria requires a marked difference between the individual's expressed/experienced gender and the gender others would assign him or her, and it must continue for at least six months. This condition may cause clinically significant distress or impairment in social, occupational or other important areas of functioning.

Gender affirming surgery is performed to change primary and/or secondary sex characteristics. For transfeminine (assigned male at birth) gender transition, surgical procedures may include genital reconstruction (vaginoplasty, penectomy, orchidectomy, clitoroplasty), breast augmentation (implants, lipofilling), and cosmetic surgery (facial reshaping, rhinoplasty, abdominoplasty, thyroid chondroplasty (laryngeal shaving), voice modification surgery (vocal cord shortening), hair transplants) (Day, 2002). For transmasculine (assigned female at birth) gender transition, surgical procedures may include mastectomy, genital reconstruction (phalloplasty, genitoplasty, hysterectomy, bilateral oophorectomy), mastectomy, and cosmetic procedures to enhance male features such as pectoral implants and chest wall recontouring (Day, 2002).

The criterion noted above for some types of genital surgeries is based on expert clinical consensus that this experience provides ample opportunity for patients to experience and socially adjust in their desired gender role, before undergoing irreversible surgery (Coleman, et al., 2022).

It is recommended that transfeminine persons undergo feminizing hormone therapy (minimum 6 months) prior to breast augmentation surgery. The purpose is to maximize breast growth in order to obtain better surgical (aesthetic) results.

In addition to hormone therapy and gender affirming surgery, psychological adjustments are necessary in affirming sex. Treatment should focus on psychological adjustment, with hormone therapy and gender affirming surgery being viewed as confirmatory procedures dependent on adequate psychological adjustment. Mental health care may need to be continued after gender affirming surgery. The overall success of treatment depends partly on the technical success of the surgery, but more crucially on the psychological adjustment of the trans identified person and the support from family, friends, employers and the medical profession.

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Nakatsuka (2012) noted that the third versions of the guideline for treatment of people with gender dysphoria (GD) of the Japanese Society of Psychiatry and Neurology recommends that feminizing/masculinizing hormone therapy and genital surgery should not be carried out until 18 years old and 20 years old, respectively. On the other hand, the sixth (2001) and the seventh (2011) versions of the standards of care for the health of transsexual, transgender, and gender non-conforming people of World Professional Association for Transgender Health (WPATH) recommend that transgender adolescents (Tanner stage 2, [mainly 12 to 13 years of age]) are treated by the endocrinologists to suppress puberty with gonadotropin-releasing hormone (GnRH) agonists until age 16 years old, after which gender-affirming hormones may be given. A questionnaire on 181 people with GID diagnosed in the Okayama University Hospital (Japan) showed that female to male (FTM) trans identified individuals hoped to begin masculinizing hormone therapy at age of 15.6 +/- 4.0 (mean +/- S.D.) whereas male to female (MTF) trans identified individuals hoped to begin feminizing hormone therapy as early as age 12.5 +/- 4.0, before presenting secondary sex characters. After confirmation of strong and persistent trans gender identification, adolescents with GD should be treated with gender-affirming hormone or puberty-delaying hormone to prevent developing undesired sex characters. These treatments may prevent transgender adolescents from attempting suicide, suffering from depression, and refusing to attend school.

Spack (2013) stated that GD is poorly understood from both mechanistic and clinical standpoints. Awareness of the condition appears to be increasing, probably because of greater societal acceptance and available hormonal treatment. Therapeutic options include hormone and surgical treatments but may be limited by insurance coverage because costs are high. For patients seeking MTF affirmation, hormone treatment includes estrogens, finasteride, spironolactone, and GnRH analogs. Surgical options include feminizing genital and facial surgery, breast augmentation, and various fat transplantations. For patients seeking a FTM gender affirmation, medical therapy includes testosterone and GnRH analogs and surgical therapy includes mammoplasty and phalloplasty. Medical therapy for both FTM and MTF can be started in early puberty, although long-term effects are not known. All patients considering treatment need counseling and medical monitoring.

Leinung and colleagues (2013) noted that the Endocrine Society's recently published clinical practice guidelines for the treatment of transgender persons acknowledged the need for further information on transgender health. These investigators reported the experience of one provider with the endocrine treatment of transgender persons over the past 2 decades. Data on demographics, clinical response to treatment, and psychosocial status were collected on all transgender persons receiving gender-affirming hormone therapy since 1991 at the endocrinology clinic at Albany Medical Center, a tertiary care referral center serving upstate New York. Through 2009, a total 192 MTF and 50 FTM transgender persons were seen. These patients had a high prevalence of mental health and psychiatric problems (over 50 %), with low rates of employment and high levels of disability. Mental health and psychiatric problems were inversely correlated with age at presentation. The prevalence of gender affirming surgery was low (31 % for MTF). The number of persons seeking treatment has increased substantially in recent years. Gender-affirming hormone therapy achieves very good results in FTM persons and is most successful in MTF persons when initiated at younger ages. The authors concluded that transgender persons seeking hormonal therapy are being seen with increasing frequency. The dysphoria present in many transgender persons is associated with significant mood disorders that interfere with successful careers. They stated that starting therapy at an earlier age may lessen the negative impact on mental health and lead to improved social outcomes.

Meyer-Bahlburg (2013) summarized for the practicing endocrinologist the current literature on the psychobiology of the development of gender identity and its variants in individuals with disorders of sex development or with transgenderism. Gender reassignment remains the treatment of choice for strong and persistent gender dysphoria in both categories, but more research is needed on the short-term and long-term effects of puberty-suppressing medications and cross-sex hormones on brain and behavior.

Note on Breast Reduction/Mastectomy and Nipple Reconstruction

The CPT codes for mastectomy (CPT codes 19303) are for breast cancer, and are not appropriate to bill for reduction mammaplasty for female to male (transmasculine) gender affirmation surgery. CPT 2020 states that “Mastectomy procedures (with the exception of gynecomastia [19300]) are performed either for treatment or

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prevention of breast cancer." CPT 2020 also states that "Code 19303 describes total removal of ipsilateral breast tissue with or without removal of skin and/or nipples (eg, nipple-sparing), for treatment or prevention of breast cancer." There are important differences between a mastectomy for breast cancer and a mastectomy for gender reassignment. The former requires careful attention to removal of all breast tissue to reduce the risk of cancer. By contrast, careful removal of all breast tissue is not essential in mastectomy for gender reassignment. In mastectomy for gender reassignment, the nipple areola complex typically can be preserved.

Some have tried to justify routinely billing CPT code 19350 for nipple reconstruction at the time of mastectomy for gender reassignment based upon the frequent need to reduce the size of the areola to give it a male appearance. However, the nipple reconstruction as defined by CPT code 19350 describes a much more involved procedure than areola reduction. The typical patient vignette for CPT code 19350, according to the AMA, is as follows: "The patient is measured in the standing position to ensure even balanced position for a location of the nipple and areola graft on the right breast. Under local anesthesia, a Skate flap is elevated at the site selected for the nipple reconstruction and constructed. A full-thickness skin graft is taken from the right groin to reconstruct the areola. The right groin donor site is closed primarily in layers."

The AMA vignette for CPT code 19318 (reduction mammaplasty) clarifies that this CPT code includes the work that is necessary to reposition and reshape the nipple to create an aesthetically pleasing result, as is necessary in female to male breast reduction. "The physician reduces the size of the breast, removing wedges of skin and breast tissue from a female patient. The physician makes a circular skin incision above the nipple, in the position to which the nipple will be elevated. Another skin incision is made around the circumference of the nipple. Two incisions are made from the circular cut above the nipple to the fold beneath the breast, one on either side of the nipple, creating a keyhole shaped skin and breast incision. Wedges of skin and breast tissue are removed until the desired size is achieved. Bleeding vessels may be ligated or cauterized. The physician elevates the nipple and its pedicle of subcutaneous tissue to its new position and sutures the nipple pedicle with layered closure. The remaining incision is repaired with layered closure" (EncoderPro, 2019). CPT code 19350 does not describe the work that that is being done, because that code describes the actual construction of a new nipple. Code 19350 is a CCI "incidental to" edit to code 19318, and, accordingly, the services of code 19350 are included in code 19318. Similarly, graft codes, such as code 15200 (full thickness skin graft) and 15877 (liposuction), are CCI "incidental to" edits to code 19318, and, accordingly, the services of graft codes, such as 15200, and liposuction codes, such as 15877, are included in code 19318.

Vulvoplasty Versus Vaginoplasty as Gender-Affirming Genital Surgery for Transgender Women

Jiang and colleagues (2018) noted that gender-affirming vaginoplasty aims to create the external female genitalia (vulva) as well as the internal vaginal canal; however, not all patients desire nor can safely undergo vaginal canal creation. These investigators described the factors influencing patient choice or surgeon recommendation of vulvoplasty (creation of the external appearance of female genitalia without creation of a neovaginal canal) and evaluated the patient's satisfaction with this choice. Gender-affirming genital surgery consults were reviewed from March 2015 until December 2017, and patients scheduled for or who had completed vulvoplasty were interviewed by telephone. These investigators reported demographic data and the reasons for choosing vulvoplasty as gender-affirming surgery for patients who either completed or were scheduled for surgery, in addition to patient reports of satisfaction with choice of surgery, satisfaction with the surgery itself, and sexual activity after surgery. A total of 486 patients were seen in consultation for trans-feminine gender-affirming genital surgery: 396 requested vaginoplasty and 39 patients requested vulvoplasty; 30 Patients either completed or are scheduled for vulvoplasty. Vulvoplasty patients were older and had higher body mass index (BMI) than those seeking vaginoplasty. The majority (63 %) of the patients seeking vulvoplasty chose this surgery despite no contraindications to vaginoplasty. The remaining patients had risk factors leading the surgeon to recommend vulvoplasty. Of those who completed surgery, 93 % were satisfied with the surgery and their decision for vulvoplasty. The authors concluded that this was the first study of factors impacting a patient's choice of or a surgeon's recommendation for vulvoplasty over vaginoplasty as gender-affirming genital surgery; it also was the first reported series of patients undergoing vulvoplasty only.

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Drawbacks of this study included its retrospective nature, non-validated questions, short-term follow-up, and selection bias in how vulvoplasty was offered. Vulvoplasty is a form of gender-affirming feminizing surgery that does not involve creation of a neovagina, and it is associated with high satisfaction and low decision regret.

Autologous Fibroblast-Seeded Amnion for Reconstruction of Neo-vagina in Transfeminine Reassignment Surgery

Seyed-Forootan and colleagues (2018) stated that plastic surgeons have used several methods for the construction of neo-vaginas, including the utilization of penile skin, free skin grafts, small bowel or recto-sigmoid grafts, an amnion graft, and cultured cells. These researchers compared the results of amnion grafts with amnion seeded with autograft fibroblasts. Over 8 years, these investigators compared the results of 24 male-to-female transsexual patients retrospectively based on their complications and levels of satisfaction; 16 patients in group A received amnion grafts with fibroblasts, and the patients in group B received only amnion grafts without any additional cellular lining. The depths, sizes, secretions, and sensations of the vaginas were evaluated. The patients were monitored for any complications, including over-secretion, stenosis, stricture, fistula formation, infection, and bleeding. The mean age of group A was 28 ± 4 years and group B was 32 ± 3 years. Patients were followed-up from 30 months to 8 years (mean of 36 ± 4) after surgery. The depth of the vaginas for group A was 14 to 16 and 13 to 16 cm for group B. There was no stenosis in neither group. The diameter of the vaginal opening was 34 to 38 mm in group A and 33 to 38 cm in group B. These researchers only had 2 cases of stricture in the neo-vagina in group B, but no stricture was recorded for group A. All of the patients had good and acceptable sensation in the neo-vagina; 75 % of patients had sexual experience and of those, 93.7 % in group A and 87.5% in group B expressed satisfaction. The authors concluded that the creation of a neo-vaginal canal and its lining with allograft amnion and seeded autologous fibroblasts is an effective method for imitating a normal vagina. The size of neo-vagina, secretion, sensation, and orgasm was good and proper. More than 93.7 % of patients had satisfaction with sexual intercourse. They stated that amnion seeded with fibroblasts extracted from the patient's own cells will result in a vagina with the proper size and moisture that can eliminate the need for long-term dilatation. The constructed vagina has a 2-layer structure and is much more resistant to trauma and laceration. No cases of stenosis or stricture were recorded. Level of Evidence = IV. These preliminary findings need to be validated by well-designed studies.

Pitch-Raising Surgery in Transfeminine Persons

Van Damme and colleagues (2017) reviewed the evidence of the effectiveness of pitch-raising surgery performed in male-to-female transsexuals. These investigators carried out a search for studies in PubMed, Web of Science, Science Direct, EBSCOhost, Google Scholar, and the references in retrieved manuscripts, using as keywords "transsexual" or "transgender" combined with terms related to voice surgery. They included 8 studies using cricothyroid approximation, 6 studies using anterior glottal web formation, and 6 studies using other surgery types or a combination of surgical techniques, leading to 20 studies in total. Objectively, a substantial rise in post-operative fundamental frequency was identified. Perceptually, mainly laryngeal web formation appeared risky for decreasing voice quality. The majority of patients appeared satisfied with the outcome. However, none of the studies used a control group and randomization process. The authors concluded that future research needs to investigate long-term effects of pitch-raising surgery using a stronger study design.

Azul and associates (2017) evaluated the currently available discursive and empirical data relating to those aspects of trans-masculine people's vocal situations that are not primarily gender-related, and identified restrictions to voice function that have been observed in this population, and made suggestions for future voice research and clinical practice. These researchers conducted a comprehensive review of the voice literature. Publications were identified by searching 6 electronic databases and bibliographies of relevant articles. A total of 22 publications met inclusion criteria. Discourses and empirical data were analyzed for factors and practices that impact on voice function and for indications of voice function-related problems in trans-masculine people. The quality of the evidence was appraised. The extent and quality of studies investigating trans-masculine people's voice function was found to be limited. There was mixed evidence to

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suggest that trans-masculine people might experience restrictions to a range of domains of voice function, including vocal power, vocal control/stability, glottal function, pitch range/variability, vocal endurance, and voice quality. The authors concluded that more research into the different factors and practices affecting trans-masculine people's voice function that took account of a range of parameters of voice function and considered participants' self-evaluations is needed to establish how functional voice production can be best supported in this population.

Facial Feminization Surgery

Raffaini and colleagues (2016) stated that gender dysphoria refers to the discomfort and distress that arise from a discrepancy between a person's gender identity and sex assigned at birth. The treatment plan for gender dysphoria varies and can include psychotherapy, hormone treatment, and gender affirmation surgery, which is, in part, an irreversible change of sexual identity. Procedures for transformation to the female sex include facial feminization surgery, vaginoplasty, clitoroplasty, and breast augmentation. Facial feminization surgery can include forehead re-modeling, rhinoplasty, mentoplasty, thyroid chondroplasty, and voice alteration procedures. These investigators reported patient satisfaction following facial feminization surgery, including outcome measurements after forehead slippage and chin re-modeling. A total of 33 patients between 19 and 40 years of age were referred for facial feminization surgery between January of 2003 and December of 2013, for a total of 180 procedures. Surgical outcome was analyzed both subjectively through questionnaires administered to patients and objectively by serial photographs. Most facial feminization surgery procedures could be safely completed in 6 months, barring complications. All patients showed excellent cosmetic results and were satisfied with their procedures. Both frontal and profile views achieved a loss of masculine features. The authors concluded that patient satisfaction following facial feminization surgery was high; they stated that the reduction of gender dysphoria had psychological and social benefits and significantly affected patient outcome. The level of evidence of this study was IV.

Morrison and associates (2018) noted that facial feminization surgery encompasses a broad range of cranio-maxillofacial surgical procedures designed to change masculine facial features into feminine features. The surgical principles of facial feminization surgery could be applied to male-to-female transsexuals and anyone desiring feminization of the face. Although the prevalence of these procedures is difficult to quantify, because of the rising prevalence of transgenderism (approximately 1 in 14,000 men) along with improved insurance coverage for gender-confirming surgery, surgeons versed in techniques, outcomes, and challenges of facial feminization surgery are needed. These researchers appraised the current facial feminization surgery literature. They carried out a comprehensive literature search of the Medline, PubMed, and Embase databases was conducted for studies published through October 2014 with multiple search terms related to facial feminization. Data on techniques, outcomes, complications, and patient satisfaction were collected. A total of 15 articles were selected and reviewed from the 24 identified, all of which were either retrospective or case series/reports. Articles covered a variety of facial feminization procedures. A total of 1,121 patients underwent facial feminization surgery, with 7 complications reported, although many articles did not explicitly comment on complications. Satisfaction was high, although most studies did not use validated or quantified approaches to address satisfaction. The authors concluded that facial feminization surgery appeared to be safe and satisfactory for patients. These researchers stated that further studies are needed to better compare different techniques to more robustly establish best practices; prospective studies and patient-reported outcomes are needed to establish quality-of-life (QOL) outcomes for patients.

In a systematic review, Gorbea et al (2021) provided a portrait of gender affirmation surgery (GAS) insurance coverage across the U.S., with attention to procedures of the head and neck. State policies on transgender care for Medicaid insurance providers were collected for all 50 states. Each state's policy on GAS and facial gender affirmation surgery (FGAS) was examined. The largest medical insurance companies in the U.S. were identified using the National Association of Insurance Commissioners Market Share report. Policies of the top 49 primary commercial medical insurance companies were examined. Medicaid policy reviews found that 18 states offer some level of gender-affirming coverage for their patients, but only 3 include FGAS (17 %); 13 states prohibit Medicaid coverage of all transgender surgery, and 19 states have no published gender-affirming medical care coverage policy; 92 % of commercial medical insurance providers had a published policy on GAS coverage. Genital reconstruction was described as a medically necessary

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aspect of transgender care in 100 % of the commercial policies reviewed; 93 % discussed coverage of FGAS, but 51 % considered these procedures cosmetic. Thyroid chondroplasty (20 %) was the most commonly covered FGAS procedure. Mandibular and frontal bone contouring, rhinoplasty, blepharoplasty, and facial rhytidectomy were each covered by 13 % of the medical policies reviewed. The authors concluded that while certain surgical aspects of gender-affirming medical care are nearly ubiquitously covered by commercial insurance providers, FGAS is considered cosmetic by most Medicaid and commercial insurance providers. Level of Evidence = V.

Hohman and Teixeira (2022) stated that with respect to gender affirmation procedures for the face, the majority of interventions will occur in patients transitioning from male to female, i.e., transgender women. While there are slightly more transgender women than transgender men in the population (33 % transgender women, 29 % transgender men, 35 % non-binary, 3 % cross-dressers, according to the USTS), the reason that more females require surgery than males is that testosterone therapy typically produces enough changes in secondary sex characteristics of the face (growth of facial hair, thickening of the skin, increase in frontal bossing, lowering of the voice, etc.) that surgery is not necessary . In some cases, placement of implants or fat transfer can increase volume in the lower 1/3 of the face and contribute to masculinization. Still, the primary area of focus for facial feminization is generally the upper 1/3. Feminization of the upper 1/3 of the face often requires several techniques to be applied in combination: The advancement of the hairline, hair transplantation, brow-lifting, and reduction of frontal bossing or "frontal cranioplasty". While the advancement of a scalp flap, hair transplant, and pretrichial brow-lifting are commonly employed cosmetic surgery interventions, frontal cranioplasty bears special consideration. Several methods of reducing the brow's prominence are often described as type 1, 2, and 3 frontal cranioplasties. Type 1 cranioplasty reduces the supra-orbital ridge's protrusion, usually using a drill, including decreasing the thickness of the anterior table of the frontal sinus. This technique is the simplest, but it is only effective in patients with either a very thick anterior frontal sinus table or an absent pneumatized frontal sinus. Type 2 cranioplasty involves augmentation of the forehead's convexity using bone cement or methyl methacrylate in addition to a reduction of the supra-orbital ridge with a drill. Type 3 cranioplasty is advocated by many prominent facial feminization surgeons and consists of removal of the anterior table of the frontal sinus, thinning of the bone flap, and replacement of that bone onto the frontal sinus but in a more recessed position, in addition to a reduction of the remainder of the supra-orbital ridge. An alternative to removal and recession of the frontal sinus's anterior table is to thin the bone with a drill and then fracture it in a controlled fashion to produce the desired contour, which is also performed routinely by some authors.

Forehead Feminization Cranioplasty

Eggerstedt and colleagues (2020) stated that forehead feminization cranioplasty (FFC) is an important component of gender-affirming surgery and has become increasingly popular in recent years. However, there is little objective evidence for the procedure's safety and clinical impact via patient-reported outcome measures (PROMs). In a systematic review, these researchers determined what complications are observed following FFC, the relative frequency of complications by surgical technique, and what impact the procedure has on patient's QOL. They carried out database searches in PubMed/Medline, Scopus, CINAHL, Cochrane CENTRAL, Cochrane Database of Systematic Reviews, and PsycINFO. The search terms included variations of forehead setback/FFC. Both controlled vocabularies (i.e., MeSH and CINAHL's Suggested Subject Terms) and keywords in the title or abstract fields were searched. Two independent reviewers screened the titles and abstracts of all articles; and 2 independent surgeon reviewers examined the full text of all included articles, and relevant data points were extracted. Main outcomes and measures included complications and complication rate observed following FFC. Additional outcome measures were the approach used, concurrent procedures carried out, and the use and findings of a PROM. A total of 10 articles describing FFC were included, entailing 673 patients. The overall pooled complication rate was 1.3 %; PROMs were used in 50 % of studies, with no standardization among studies. The authors concluded that complications following FFC were rare and infrequently required reoperation. Moreover, these researchers stated that further studies into standardized and validated PROMs in facial feminization patients are needed. Level of Evidence = III.

Hand Feminization and Masculinization

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Lee and colleagues (2021) noted that anatomical characteristics that are incongruent with an individual's gender identity can cause significant gender dysphoria. Hands exhibit prominent dimorphic sexual features, but despite their visibility, there are limited studies examining gender affirming procedures for the hands. These researchers examined the anatomical features that define feminine and masculine hands, the surgical and non-surgical approaches for feminization and masculinization of the hand; and adapted established aesthetic hand techniques for gender affirming care. They carried out a comprehensive database search of PubMed, Embase OVID and SCOPUS to identify articles on the characterization of feminine or masculine hands, hand treatments related to gender affirmation, and articles related to techniques for hand feminization and masculinization in the non-transgender population. From 656 possibly relevant articles, 42 met the inclusion criteria for the current literature search. There is currently no medical literature specifically examining the surgical or non-surgical options for hand gender affirmation. The available techniques for gender affirming procedures discussed in this paper were appropriated from those more commonly used for hand rejuvenation. The authors concluded that there is very little evidence addressing the options for transgender individuals seeking gender affirming procedures of the hand. These researchers stated that although established procedures used for hand rejuvenation may be employed in gender affirming care, further study is needed to determine relative salience of various hand features to gender dysphoria in transgender patients of various identities, as well as development of novel techniques to meet these needs. Level of Evidence = III.

Peritoneal Pull-Through Technique Vaginoplasty in Neovagina Construction in Gender-Affirming Surgery

Tay and Lo (2022) reviewed the application, effectiveness and outcomes of a novel surgical technique, peritoneal pull-through technique vaginoplasty, in gender-affirming surgery. Specific outcome parameters included healing time, depth of cavity achieved,) alleviation of dysphoria, and morbidity of the surgery. These researchers carried out a systematic review according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement and PROSPERO registration obtained before commencement. A search was performed in OVID Medline, Embase, Willey Online Library and PubMed. Specialty-related journals, grey literature and reference lists of relevant articles were manually searched. From 476 potentially relevant articles, 12 articles were analyzed; and the publications were all level 4 or level 5 evidence. Healing times were poorly reported or often not mentioned. A total of 8 authors reported neovagina cavity depth of at least 13 cm and good patient satisfaction. Alleviation of dysphoria was not discussed by any of the publications and only 6reported complications. Average follow-up ranged from 6 weeks to 14.8 months. The authors concluded that the use of peritoneal pull-through vaginoplasty in gender-affirming surgery is promising and novel; however, there is a paucity of data. These investigators stated that further research and longer-term data are needed to examine the safety and effectiveness of this technique including stabilization of vaginal depth, later morbidity and complications. Patients seeking this surgery overseas should be informed of the potential difficulties they may face.

Urethral Complications and Outcomes in Transgender Men

Hu et al (2022) noted that urologic problems, such as urethral fistulas and strictures, are among the most frequent complications following phalloplasty. Although many studies have reported successful phalloplasty and urethral reconstruction with reliable outcomes in transgender men; so far, no method has become standardized. These researchers examined the reports on urological complications and outcomes in transgender men with respect to various types of urethral reconstruction. They carried out a comprehensive literature search of PubMed, Scopus, and Google Scholar databases for studies related to phalloplasty in transsexuals. Data on various phallic urethral techniques, urethral complications, and outcomes were collected and analyzed using the random-effects model. A total of 21 studies (1,566 patients) were included: 8 studies (1,061 patients) on "tube-in-tube", 9 studies (273 patients) on "prelaminated flap, and 6 studies (221 patients) on "second flap". Compared with the tube-in-tube technique, the pre-laminated flap was associated with a significantly higher urethral stricture/stenosis rate; however, there was no difference between the pre-laminated flap and the 2nd flap techniques. For all phalloplasty patients, the pooled rate of urethral fistula or stenosis was 48.9 %, the rate of the ability to void while standing was 91.5 %, occurrence rate of tactile or erogenous sensation was 88 %, the prosthesis complication rate was 27.9 %, and patient-

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reported satisfactory outcome rate was 90.5 %. The authors concluded that urethral reconstruction with a pre-laminated flap was associated with a significantly higher urethral stricture rate and increased need of revision surgery compared with that observed using a skin flap. Overall, most patients were able to void while standing and were satisfied with the outcomes.

Facial Gender Affirming Procedures

Oles et al (2022) stated that rapidly increasing clinical volumes of gender-affirming surgeries have stimulated a growing need for high-quality clinical research. Although some procedures have been carried out for decades, each individual procedure has limited data, necessitating synthesis of the entire literature to understand current knowledge and guide future research. These investigators carried out the 1st systematic review of all available GAS publications across all procedures to evaluate both outcomes reported in the literature as well as the methods used for outcome assessment. This review was conducted following PRISMA guidelines to identify all outcomes measures in GAS cohorts, including PCOs, complications, and functional outcomes. Outcome data were pooled to examine currently reported complication, satisfaction, and other outcome rates. Overall, 15,186 references were identified, 4,162 papers advanced to abstract review, and 1,826 underwent full-text review. After review, there were 406 GAS cohort publications. Of non-genitoplasty titles, 35 were mastectomy, 6 mammoplasty, 21 facial feminization, and 31 voice/cartilage. Although 59.1 % of non-genitoplasty papers addressed PCOs in some form, only 4.3 % used instruments partially-validated in transgender patients. Overall, data were reported heterogeneously and were biased towards high-volume centers. The authors concluded that this review represented the most comprehensive review of GAS literature. By aggregating all previously utilized measurement instruments, this study offered a foundation for discussions regarding current methodologic limitations and what dimensions must be included in assessing surgical success. These investigators have assembled a comprehensive list of outcome instruments; this offered an ideal starting basis for emerging discussions between patients and providers regarding deficiencies which new, better instruments and metrics must address. The lack of consistent use of the same outcome measures and validated GAS-specific instruments represent the 2 primary barriers to high-quality research where improvement efforts should be focused.

Siringo et al (2022) noted that facial feminization surgery is composed of a variety of cranio-maxillofacial surgical procedures that are increasingly sought after by male-to-female transgender patients and by those seeking feminization of the face. Facial feminization surgery can play a prominent role in alleviating gender dysphoria. In consideration of an observed increase in gender-affirming procedures carried out in recent years, a broad knowledge base in the techniques, outcomes, and challenges of facial feminization surgery should be established by surgeons offering these procedures. In a systematic review, these investigators examined the available evidence and informed future advancements in gender-affirming surgical practice. Furthermore, they detailed a representative case to illustrate the senior author's approach to full facial feminization. These researchers carried out a comprehensive literature search of the PubMed, Embase, and Cochrane databases for studies published through June 2020 using following the search terms: "Face" OR "facial" OR "craniofacial" AND "Feminization" OR "Feminization Surgery" OR "Gender Reassignment" OR "Gender Affirming Surgery" OR "Gender Confirmation Surgery". Data on procedures, outcomes, patient age, follow-up time, complications, and patient satisfaction were collected. The data were categorized by facial thirds and then further stratified by facial feature. The search yielded 388 articles; 30 fit the inclusion criteria, and of these, 23 were included in the review. These investigators extracted primary data pertaining to 3,554 patients with an age range of 18 to 73 years who underwent 8,506 total procedures. Most of the procedures addressed the upper facial third (hairline, forehead, and brow), comprising 49.1 % of total procedures performed. Further categorization by facial feature revealed that the most commonly addressed feature was the forehead (34.6 % of procedures), followed by the nose (12.8 %) and the chin (12.2 %). In studies that used quantitative measures to gauge patient-reported outcomes, satisfaction was high. The authors concluded that facial feminization surgery appeared to be safe, whether it was conducted in a single stage or as a staged procedure. Patients reported high satisfaction and better gender congruency following facial feminization procedures. Moreover, these researchers stated that further investigation is needed to establish best surgical practice and gauge patient satisfaction beyond the length of average follow-up and determine the frequency with which adjunctive procedures are sought out.

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Coon et al (2022) stated that increasing societal acceptance of transgender individuals has resulted in broader availability of gender surgery and rapid growth in transition-related operations. Facial gender surgery aims to modify patients' facial features to be more congruent with their physical expression of gender, reducing gender dysphoria and improving QOL. Growth in research and technique evolution has not kept pace with growth in clinical volume; thus, the 1st International Facial Gender Symposium was held at Johns Hopkins University in 2019, convening surgeons who perform facial gender surgery to share ideas and examine the state of clinical evidence. These investigators reviewed the literature on facial gender surgery, they developed a search strategy for 7 electronic databases (PubMed, PsycINFO, Embase, CINAHL, Web of Science, Cochrane, and Gender Studies) through May of 2019, following the PRISMA systematic review guidelines. Based on the English language literature and clinical experience, the authors suggested guidelines for screening, management, and appropriate surgical technique for patients undergoing facial gender surgery. They highlighted facial gender surgery as a medically necessary intervention and identify shortcomings in current guidelines. The authors concluded that facial gender surgery represents a complex array of cranio-facial and soft-tissue procedures that require application of advanced skills and decision-making. Facial gender operations are not cosmetic, are medically necessary, and require development of new CPT codes specific to facial gender surgery. It is imperative to create educational programs and methods to define sufficient training for facial gender surgery surgeons. These investigators stated that research priorities include better procedural outcomes data, more quality-of-life studies, and insight into variation in both patient and procedural subgroups.

Javier et al (2022) noted that many transgender women and men undergo GAS. Existing work showed that early surgery outcomes are generally positive, suggesting high surgical satisfaction and positive QOL outcomes. Less work, however, examines these outcomes in the longer-term. These investigators carried out a systematic literature review into the longer-term (i.e., 1 year or longer) surgical satisfaction and quality of outcomes following various forms of GAS in transgender populations. Specifically, they examined research on such outcomes at least 1-year post gender-affirming chest, genital, facial, vocal cord, and Adam's apple removal surgeries. Studies were identified via Google Scholar, PsycINFO, Scopus, and PubMed databases, as well as through Google Scholar search alerts. These researchers considered all studies published until October 2021; 2 reviewers extracted data from suitable studies using Covidence. Both reviewers also independently examined the identified studies' risk of bias and strength of evidence. A total of 79 low quality (e.g., small sample sizes, lack of control/comparison groups) studies suggested that most transgender patients were satisfied with surgical outcomes when assessed at least 1-year post-surgery. Low quality research also indicated that transgender women and men typically reported positive psychological and sexual well-being post-surgery, and similar well-being outcomes as those who have not had surgery. The authors concluded that to the best of their knowledge, this literature review was the first to critically summarize and evaluate all published studies on the longer-term QOL outcomes following chest, genital, facial, voice and Adam's apple removal surgeries for transgender women and men. These investigators stated that while the results suggested promising surgical satisfaction and QOL outcomes following surgery, many studies only drew on small samples, and most studies did not allow for causal conclusions. Furthermore, few studies have compared surgical outcomes between transgender women and men. These investigators offered concrete suggestions for future research.

La Padula et al (2023) noted that facial feminization surgery (FFS) is often the first procedure requested by patients wishing to undergo GAS. These researchers examined the applicability and effectiveness of deep plane facelifts in FFS. They carried out a prospective study that included patients who requested a deep plane facelift as a stand-alone procedure to achieve a more feminine facial appearance as the only procedure of FFS. Subjects underwent deep plane facelifts to achieve a more feminine oval face shape and increased tissue projection of the zygomatic-malar region. To examine the effectiveness of the procedure and patient satisfaction, the Face-Q scales, Face and Neck lift Objective Photo-Numerical Assessment Scale, the Satisfaction With Life Scale, and the Subjective Happiness Scale were applied pre-operatively and 1 year after surgery. A total of 36 patients were included in the study. A statistically significant difference ($p < 0.005$) was observed between pre- and post-operative scores. The re-positioning of the malar fat pads increased the malar volume, providing a more oval overall shape of the face, which is typically feminine. No major complications were observed. The authors concluded that despite their encouraging results, new studies with a larger sample of patients are needed to support the benefits of the deep plane facelift as part of FFS to elevate this technique from an ancillary to a routine procedure for patients undergoing GAS.

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Hoffman et al (2024) stated that facial masculinization surgery (FMS) is increasingly popular among cisgender and transgender men. The benefits of FMS are focused on facial identity and have been proven to decrease gender dysphoria in this population. Previous research showed increasing interest in GAS and facial feminization surgery; however, the prevalence of FMS has not been explored. It is difficult to find these data based on surgical records alone because institutions do not have standardized methods of reporting and lack publications in the field. These investigators analyzed public interest in FMS by using worldwide Google Trends to quantify these trends. They carried out a worldwide Google Trends search from January 1, 2008, to December 31, 2022, using terms focused on FMS. Then, search terms were analyzed for non-facial masculinization procedures and were aggregated. Lastly, a PubMed search was carried out for the terms "transgender" and "facial masculinization" from January 1, 2008, to December 31 , 2022, to compare publication rates. The data showed an increasing interest in FMS via Google search trends since the year 2008. A similar trend was demonstrated for non-FMS gender-affirming terms. PubMed analysis showed "transgender" medicine publishing rates were approximately 39.65 times greater than "facial masculinization" publishing rates, although "facial masculinization" medicine did produce a positive trend over the study period of approximately 4 publications per year. The medical literature on transgender surgeries rapidly outpaces publications specifically focusing on FMS. The authors concluded that this study showed increasing interest in GAS over time, especially FMS. These increasing trends should encourage greater scientific exploration of FMS and research to properly quantify and assess surgical outcomes in this special population. These investigators stated that additional educational interventions for both the general public and medical providers, to increase awareness of unique challenges that impact this community and highlight changes in healthcare coverage over time, should be created to keep pace with increasing patient demand and address the physical, systemic, and psychosocial issues faced by individuals who identify as transgender.

An UpToDate review on “Gender-affirming surgery: Male to female” (Ferrando, 2024) states that “Nongenital/nonbreast surgeries include a range of feminizing surgeries. While facial feminization and thyroid cartilage reduction are discussed here, other feminizing procedures include body contouring (e.g., liposuction) and body implants (e.g., buttock implants). WPATH has published recommended criteria that need to be met by patients considering and clinicians performing these types of surgeries. Patients should be fully counseled and informed of the risks and benefits prior to proceeding with surgery”.

Furthermore, an UpToDate review on “Gender-affirming surgery: Female to male” (Ferrando et al, 2024) does not mention facial gender affirming surgery.

Nerve Surgery / Neurotization at Gender-Affirming Breast Surgery

Rochlin et al (2020) noted that female-to-male mastectomy often renders the chest skin and nipple-areolar complex (NAC) insensate. These researchers proposed a new technique of preserving the intercostal nerves (ICN) and using them to re-innervate the NAC following mastectomy. They carried out a prospective analysis of transmasculine patients who underwent female-to-male mastectomy. The technique entailed dissecting out the lateral ICN to length and performing a neurorrhaphy to nerve stumps at the base of the NAC. Sensory outcomes, as assessed with Semmes-Weinstein monofilaments, were compared to a cohort of patients who underwent mastectomy without neurotization. A total of 10 patients with a mean age of 17.5 years (range of 16 to 19 years) underwent mastectomy. The final follow-up was a mean of 15.4 ± 4.3 months for the treated group and 40.7 ± 12.9 months for the control group. Compared to control patients, treated patients had significant improvement in sensation at the nipple ($p \leq 0.0002$), areola ($p = 0.0001$), and peripheral breast skin ($p = 0.0001$). For treated patients, there was no statistically significant difference in sensation between pre-operative and post-operative sensation in all tested areas at final follow-up. The authors concluded that this proof of concept (POC) study suggested that immediate reinnervation of the NAC following mastectomy enhanced recovery of NAC sensation in patients undergoing female-to-male mastectomy and may be further generalized to women undergoing post-mastectomy breast reconstruction. These researchers stated that immediate targeted NAC reinnervation has the potential to herald the “next frontier” in breast reconstruction, while also improving outcomes in transmasculine patients undergoing gender-affirming mastectomy.

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The authors stated that the drawbacks of this study included the small study size (n = 10). They focused on objective sensory outcomes based on monofilament and temperature testing, although subjective outcomes, such as sexual arousal and self-esteem, are also important clinical endpoints. Furthermore, the placebo effect was a potential bias, as knowledge of neurotization may have impacted the treated group's perception of sensation. To avoid this potential bias, future studies could employ a prospective, single-blind, randomized-controlled study design comparing transmasculine patients undergoing mastectomy with and without neurotization to evaluate sensory outcomes. Alternatively, future studies could examine differential sensory recovery of the chest in patients receiving unilateral nipple neurotization. Comparison of sensory recovery in patients who undergo nipple-sparing mastectomy (NSM) versus free nipple grafting (FNG) is also an interesting line of inquiry that can be addressed in future investigations.

Deptula and Nguyen (2021) stated that NSM is commonly performed when oncologically feasible. Although the NAC is preserved, patients are left with significantly altered nipple sensation after mastectomy. To optimize breast reconstruction outcomes, the NAC as well as its sensation should ideally be preserved. These investigators examined outcomes of NAC neurotization in patients undergoing mastectomy for both gender-affirmation and oncologic indications. They carried out a retrospective chart review on patients who underwent nipple reinnervation procedures from 2016 to 2019. Two groups of patients were identified who underwent nipple reinnervation during gender-affirming mastectomy or NSM and immediate autologous reconstruction. Patient demographic information and sensory assessments were recorded. Sensation was assessed for nipple, areola and peripheral breast skin (2 cm from the areola edge) sensation using Semmes-Weinstein monofilaments. A total of 10 patients underwent gender-affirming mastectomies with NAC reinnervation. NAC neurotization patients reported significantly better sensation at the nipple ($p = 0.0001$), areola ($p = 0.0001$) and surrounding breast skin ($p = 0.0001$) compared to a control group; 7 patients underwent reinnervation group after NSM and autologous reconstruction. NAC neurotization resulted in higher levels of sensation in the reinnervation with flap group compared a control group at the nipple (3.9 versus 4.9, $p = 0.04$) and areola (4.84 versus 5.68, $p = 0.04$). The presence of neuroma or hypersensitivity of the NAC were not observed in any patient undergoing neurotization (0 %). The authors concluded that the NAC has a valuable sensory function that is worth preserving after NSM. NAC neurotization is a viable technique in 2 patient populations: transmasculine patients undergoing gender-affirming mastectomy and breast cancer patients undergoing NSM and immediate autologous reconstruction. Moreover, these researchers stated that further study is needed to understand patient satisfaction and QOL outcomes with nipple reinnervation.

The author stated that the main drawback of this study was its retrospective design. While these investigators managed to identify control groups, they were not an equivalent substitution for a randomized controlled trial (RCT) to compare outcomes. Another drawback was the relatively small sample size (n = 10). The lack of statistical significance in some of these findings may have been related to an under-powered study. This trial was focused on the sensation of the NAC after neurotization. These researchers stated that further study may include a detailed analysis of sensation quadrate specific mastectomy skin as well as the effect of multiple coaptations when surgically feasible. While the authors employed objective and repeatable assessments of sensation, they did not carry out patient-reported outcome assessments. As these researchers continue to refine their NAC neurotization techniques, efforts will focus on patient reported outcome assessments using validated instruments such as the BREAST-Q and the BODY-Q chest module.

Troung et al (2023) recently described targeted nipple reinnervation (TNR) during female-to-male gender-affirming mastectomy with FNG using either direct nerve coaptation or nerve allograft. The objectives of TNR are to improve sensation (including erogenous sensation) and prevent numbness, paresthesia, chronic pain, and phantom sensation. These investigators described their modified technique, which has evolved to use autologous ICN branches as donor nerves for re-innervation if direct nerve coaptation cannot be achieved. During TNR, the T3 to T5 sensory branches are preserved and coapted to the repositioned NAC. In patients with donor nerves that were inadequate in length to allow for direct coaptation, autologous ICN branches were not used for coaptation (branches present along the chest wall that would otherwise be lost) or one of the T3 to T5 branches were harvested. An end-to-end nerve repair between the autograft and donor nerves was performed, and the donor nerve/autograft complex was coapted to the NAC. Targeted muscle re-innervation was carried out after autograft harvest to prevent neuroma formation. TNR with ICN autograft is technically feasible in female-to-male gender-affirming mastectomy with FNG when direct coaptation is not

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possible. Chest re-innervation using autologous ICN branches as donor nerves is another option for re-innervation when the nerves are too short for direct coaptation. The authors stated that because the collection of long-term data is ongoing, the effectiveness of NAC re-innervation using the new technique will be described in a future publication. Moreover, these researchers stated that long-term follow-up data are needed to examine the sensory outcome and patient satisfaction following autograft nerve repair. Furthermore, it will be important to better understand the post-operative time to sensory return and compare the effectiveness of direct coaptation versus autograft and allografts for TNR.

Remy et al (2025) stated that TNR is a novel technique to preserve and reconstruct ICN to improve post-operative sensation; however, there is little evidence on relevant anatomy and outcomes. These investigators analyzed the anatomy and sensory outcomes of TNR in gender-affirming double incision mastectomy with FNG. A total of 25 patients were prospectively enrolled. Data included demographics, surgical technique, and axon/fascicle counts. Quantitative sensory evaluation using monofilaments and qualitative patient reported questionnaires were completed pre-operatively, and at 1, 3, 6, 9, and 12 months post-operatively. A total of 50 mastectomies were carried out. Per mastectomy, the median number of ICN found and used was 2 (1 to 5). Axon and fascicle counts were not significantly different between ICN branches ($p > 0.05$). BMI of 30 kg/m2 or greater and mastectomy weight of 800 g or greater were associated with significantly worse pre-operative sensation ($p < 0.05$). Compared to pre-operative values, NAC sensation was worse at 1 month ($p < 0.01$), comparable at 3 months ($p > 0.05$), and significantly better at 12 months ($p < 0.05$) post-operatively. Chest sensation was comparable to the pre-operative measurements at 1 and 3 months ($p > 0.05$), and significantly better at 12 months ($p < 0.05$) post-operatively. NAC sensation was significantly better when direct coaptation was performed compared to use of allograft only ($p < 0.05$), and with direct coaptation of 2 or more branches compared to direct coaptation of 1 branch ($p < 0.05$). All patients reported return of nipple and chest sensation at 1 year post-operatively and 88 % reported return of some degree of erogenous sensation. The authors concluded that TNR allowed for restoration of NAC and chest sensation within 3 months post-operatively; and the use of multiple ICN branches and direct coaptation resulted in the best sensory outcomes.

Remy et al (2024b) analyzed the quantitative and qualitative sensory outcomes following gender-affirming mastectomy (GAM) with TNR. A total of 60 patients were prospectively enrolled -- 30 subjects undergoing GAM with TNR were compared to 30 matched controls who underwent GAM without TNR and 30 cis-males who did not undergo surgery. Quantitative sensory testing including monofilament testing, temperature detection threshold, pressure pain threshold, 2-point discrimination, pin-prick, and vibration, as well as patient-reported questionnaires were completed pre-operatively and at 1, 3, 6, 9 and 12 months post-operatively. Monofilament values, cold and warm detection thresholds, pin-prick and pressure detection threshold were comparable to pre-operative values in subjects ($p > 0.05$) but remained significantly worse in controls ($p < 0.05$). Vibration and 2-point discrimination remained similar to pre-operative values in both subjects and controls ($p > 0.05$). Subjects reported significantly better nipple, temperature and erogenous sensation compared to controls ($p < 0.05$). Chest pain was significantly higher in controls versus subjects until 6 months post-operatively ($p < 0.05$). Phantom sensation and phantom pain were significantly higher in controls compared to subjects ($p < 0.05$). Nipple hypersensitivity was significantly higher in subjects compared to controls until 6 months post-operatively ($p < 0.05$). The authors concluded that restoration of mechanical (light touch and pressure), temperature and pain detection, as well as erogenous sensation were significantly improved with TNR, which was associated with significantly less phantom sensation and chest pain. Patients should be counselled on transient NAC hypersensitivity.

Current UpToDate reviews on “Gender-affirming surgery: Female to male” (Ferrando et al, 2024), and “Gender-affirming surgery: Male to female” (Ferrando, 2024) do not mention nerve grafting/neurotization for breast reconstruction.

Facial Recognition Neural Networks for Confirmation of Success of Facial Feminization Surgery

Chen et al (2020) noted that male-to-female transgender patients desire to be identified, and treated, as female, in public and social settings. Facial feminization surgery entails a combination of highly visible

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changes in facial features. To study the effectiveness of facial feminization surgery, these investigators examined pre-operative/post-operative gender-typing using facial recognition neural networks. In this study, standardized frontal and lateral view pre-operative and post-operative images of 20 male-to-female patients who completed hard-tissue and soft-tissue facial feminization surgery procedures were used, along with control images of unoperated cisgender men and women (n = 120 images). Four public neural networks trained to identify gender based on facial features analyzed the images. Correct gender-typing, improvement in gender-typing (pre-operatively to post-operatively), and confidence in femininity were analyzed. Cisgender male and female control frontal images were correctly identified 100 % and 98 %of the time, respectively. Pre-operative facial feminization surgery images were misgendered 47 % of the time (recognized as male) and only correctly identified as female 53 % of the time. Post-operative facial feminization surgery images were gendered correctly 98 % of the time; this was an improvement of 45 %. Confidence in femininity also improved from a mean score of 0.27 before facial feminization surgery to 0.87 after facial feminization surgery. The authors concluded that in the 1st study of its kind, facial recognition neural networks demonstrated improved gender-typing of transgender women from pre-operative facial feminization surgery to post-operative facial feminization surgery. This showed the effectiveness of facial feminization surgery by artificial intelligence (AI) methods. Level of Evidence = IV.

In a commentary on the afore-mentioned study by Chen et al (2020), Zuo and Forrest (2021) noted some of the confounding variables, such as hormonal therapy and use of make-up, which may affect the distinction between male and female. Zuo and Forrest (2021) also pointed out that the study by Chen et al (2020) was a very small study (n = 20). Moreover, they noted that as suggested by Chen et al (2020), future studies could also examine the marginal impact of individual facial feminization surgery procedures on facial recognition technology performance to determine which procedures elicit the greatest effects.

In a scoping review, Ahmadi et al (2023) examined the use of ML in the facial cosmetic surgeries and procedures. Electronic search was conducted in PubMed, Scopus, Embase, Web of Science, ArXiv and Cochrane databases for the studies published until August 2022. Studies that reported the use of ML in various fields of facial cosmetic surgeries were included. The studies' risk of bias (ROB) was examined using the QUADAS-2 tool and NIH tool for before and after studies. From 848 studies, a total of 29 studies were included and categorized in 5 groups based on the objective of the studies: outcome evaluation (n = 8), face recognition (n = 7), outcome prediction (n = 7), patient concern evaluation (n = 4) and diagnosis (n = 3). A total of 16 studies used public data sets. ROB assessment using QUADAS-2 tool revealed that 6 studies were at low ROB, 5 studies were at high ROB, and others had moderate ROB. All studies assessed with NIH tool showed fair quality. In general, all studies showed that using ML in the facial cosmetic surgeries was accurate enough to benefit both surgeons and patients. The authors concluded that using ML in the field of facial cosmetic surgery is a novel method and needs further studies, especially in the fields of diagnosis and treatment planning. Due to the small number of studies and the qualitative analysis conducted, these researchers could not draw a general conclusion regarding the impact of ML in the field of facial cosmetic surgery. Level of Evidence = IV.

Nerve Grafting for Gender Affirming Breast Surgery

Gfrerer et al (2022) noted that restoration of breast sensation has become an important objective in autologous and implant-based breast reconstruction after cancer-related mastectomy. Although gender-affirming mastectomy with free nipple grafting (FNG) results in similar sensory deficits, chest reinnervation concepts have not been applied to this procedure. These researchers described a novel technique to reinnervate the FNG in patients undergoing double incision gender mastectomy. This technique differs from previously described reinnervation techniques in several aspects: First, the donor axon count is maximized by preserving the 3rd to 5th lateral cutaneous nerves for coaptation to the nipple areola complex. Second, the reinnervation approach varies and is based on patient anatomy. Third, the distal graft or donor nerve is split into fascicles to increase the reinnervation zone. Fourth, the split fascicles are coapted to the dermato-sensory peripheral nerve elements of the dermis. The authors concluded that chest reinnervation is technically feasible in patients undergoing double incision gender mastectomy with FNG.

Loughran et al (2024) stated that FNG makes sensory recovery challenging. Permanent decreased sensation to touch and temperature in skin-grafted skin is common. Direct neurotization of the nipple-areolar complex

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(NAC) graft has been described; however, quantitative data regarding degree of nipple reinnervation possible is unknown. These researchers quantified and qualified sensation recovery following nerve coaptation to reinnervate the NAC. Patients undergoing mastectomy for gender dysphoria from 2020 to 2022 were offered nerve allograft to restore nipple sensation. A lateral intercostal nerve was selected and coapted to allograft which was embedded beneath the nipple graft. Semmes Weinstein testing was used to evaluate nipple sensation. Assessments were made at visits of less than 1 year and over 1 year from surgery. Filaments used represented normal sensation, minor diminished sensation, diminished sensation, loss of protective sensation, and deep pressure sensation only. A total of 115 patients elected for direct neurotization. Semmes Weinstein testing was limited to 46 patients representing 46 encounters and 92 nipples in the less than 1 year group and 24 encounters and 48 nipples in the over 1 year group. Of the 92 nipples in the less than 1 year group, 17 (18.5 %) noted return of normal sensation and 37 (40.2 %) noted minor diminished or diminished sensation, indicating nerve reinnervation. There were 38 (41 %) nipples with loss of protective sensation or deep sensation only. There were 48 nipples included in the over 1 year group. Of the 48 nipples, 4 (8.3 %) noted normal sensation and 30 (62.5 %) noted minor diminished or diminished sensation, indicating nerve reinnervation. For the remaining 14 nipples, 14 (29 %) noted loss of protective sensation or deep sensation only. The authors concluded that sensory outcomes in NAC grafts used for reconstruction in patients undergoing double incision mastectomy remain poor. Sensation restoration beyond that expected from full thickness skin grafts can be achieved in the majority of patients with nerve allograft via direct neurotization.

The authors stated that this study was limited by loss to follow-up, especially in the greater than 1 year group. Despite this limitation, this study was the first of its kind to include a large patient cohort and to quantify degree of sensory recovery using Semmes Weinstein monofilament testing in patients undergoing direct neurotization of NAC graft after double incision mastectomy for gender dysphoria. All patients recovered a degree of return of sensation, with 71 % of patients demonstrating normal, minor diminished light touch or diminished light touch return of sensation after 1 year. This value is well beyond what would be expected in a full thickness skin graft. Furthermore, the authors were unable to provide a reliable control group for comparison in this study, as the vast majority of patients who elected for nerve grafting also elected for direct neurotization of the free nipple graft.

Remy et al (2025) stated that most patients undergoing breast surgery with FNG lose nipple erection (NE) function. These investigators examined the effect of nerve preservation and reconstruction with targeted nipple-areola complex reinnervation (TNR) on NE following gender-affirming mastectomy with FNG. Patients undergoing gender-affirming mastectomy with FNG were prospectively enrolled. Subjects who underwent TNR were compared with controls who did not undergo TNR. Post-operative patient-reported NE function was scored using a 4-point Likert scale. Objective NE evaluation consisted of the change in areola circumference and nipple height following cold application using a thermal device and 3-dimensional (3D) imaging. A total of 20 patients (11 subjects and 9 controls) with comparable age, body mass index (BMI), and mastectomy weight were included. At an average follow-up of 16.8 (± 7.0) months, significantly more subjects reported NE than controls (72.8 % versus 38.9 %, $p = 0.03$), with a higher median NE score (3 [range of 1 to 4] versus 1 [range of 1 to 2], $p = 0.0005$). Following cold application, subjects had a greater mean reduction in areola circumference (-4.16 ± 3.3 versus -1.67 ± 1.9 mm, $p = 0.02$) and a greater mean increase in nipple height (+0.86 ± 0.8 versus +0.37 ± 0.3 mm, $p = 0.04$) compared with controls. Improved patient-reported NE function correlated with better cold detection thresholds ($p = 0.01$). The authors concluded that TNR was associated with improved patient-reported and objective NE following gender-affirming mastectomy. These investigators stated that these findings suggested the importance of both afferent sensory and efferent autonomic intercostal nerve fiber functions in mediating NE and suggested the effectiveness of TNR in improving NE following breast surgery involving FNG.

The authors stated that the drawbacks of this trial mainly included its small sample size (n = 11 subjects) and that subjects were not blinded or randomized to the reinnervation intervention. In addition, the evaluation of NE function pre-operatively and then post-operatively at multiple follow-ups, or at least at the same follow-up period in all patients, would have provided additional information. Moreover, although age, BMI, and mastectomy were statistically comparable between both study groups, other variables which may have influenced outcomes were not analyzed, including pre-operative NAC size and surgeon experience and technique. Furthermore, patient-reported NE function did not distinguish NE in the context of cold sensation versus erogenous sensation, and objective NE function was only measured in the context of cold exposure.

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Lastly, these researchers did not examine patient pre-operative concern for NE function to understand the importance of preserving this function in patients undergoing gender-affirming mastectomy. These researchers stated that future studies should aim to stratify NE function based on different stimuli.

Frontal Contouring with Corrugator Resection

Hu et al (2024) noted that given the significance of fore-head aesthetics to facial gender identification, frontal sinus setback and brow contouring are frequent components of FFS in transgender female patients. These investigators presented a safe and reliable technique for corrugator muscle resection during FFS to provide enhanced feminization of the fore-head. This technique was carried out in 12 patients aged 27 to 58 years. Following bone reduction and sinus setback, the corrugator was located on the under-surface of the fore-head flap and dissected via the reflected soft tissue of the fore-head, with care taken to protect the supra-orbital nerve. Patient satisfaction and complications, such as infection, poor wound healing, scarring, as well as desire for revision surgery, were assessed at follow-up visits. All participants demonstrated improved feminization of the fore-head. No complications were reported intra-operatively or post-operatively, and no subjects required hospitalization beyond the expected over-night admission. No patients reported anesthesia or hypoesthesia in the distribution of the supra-orbital nerve at 6 months post-operatively. Participant reported high satisfaction with their FFS outcomes. The authors concluded that performing corrugator resection in conjunction with FFS may enhance the feminizing effects on the fore-head of frontal sinus setback and brow contouring. Moreover, these investigators stated that future studies should focus on quantifying the change attributable to corrugator resection alone. These researchers stated that this work was limited by a relatively short (6 months) follow-up period, a small patient sample (n = 12), concurrent frontal sinus setback, and the use of non-standardized measures of patient satisfaction.

Intercostal Nerve Transfer for Sensation Preservation in Gender-Affirming Subcutaneous Mastectomy

Lin et al (2025) noted that chest-wall contouring surgery is an important step in the transitional journey of many transgender individuals who choose to undergo masculinization procedures. Traditional gender-affirming chest surgery does not include reinnervation of the NAC or the mastectomy skin flaps and risks partial or complete denervation of these areas. Efforts to restore sensation have emerged resulting in various sensory nerve transfer techniques including direct neurorrhaphy to the re-positioned NAC or to underlying dermato-sensory elements. These researchers described a novel technique in which they carried out a nerve transfer from the anterior lateral branch of the 4th intercostal nerve (of the breast parenchyma to be resected) to the anterior lateral branch of the 3rd or 2nd intercostal nerve in the mastectomy skin flap. There has been no increased risk of complications, and results so far are promising. The authors concluded that sensory transfer from the 4th to the 3rd intercostal nerve is a novel technique to restore chest sensation in gender-affirming mastectomy. The technique is safe, does not require additional or extended incisions, and has no effect on aesthetic outcomes. Moreover, these researchers stated that further investigations with longer follow-up are needed to better quantify outcomes; they noted that currently, the outcomes are promising, suggesting this is a viable proof of concept.

The authors stated that limitations of this trial included the lack of precise mapping of chest-wall sensory input, potentially affecting the correlation of erogenous zones with nipple grafts. However, erogenous zones do vary among individuals, and specific geographic mapping may not be as relevant if erogenous sensation can be preserved. These investigators stated that future directions include the potential to restore sensation in oncological mastectomies. The current fore-front of procedures focuses on preserving sensation, in some cases by preserving the superficial anterior branch of the intercostal nerve when oncologically possible. In cases where the 4th intercostal branch cannot be spared, options such as nerve grafting or performing a sensory nerve transfer of the 3rd intercostal to the 4th intercostal nerve.

Wendler Glottoplasty for Gender Affirming Voice Surgery

In a prospective study, Aires et al (2023) examined the effect of Wendler glottoplasty (WG) on voice feminization, voice quality and voice-related QOL. This trial included transgender women who underwent

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WG. Acoustic analysis of the voice included assessment of fundamental frequency, maximum phonation time formant frequencies (F1 and F2), frequency range, jitter and shimmer. Voice quality was blindly evaluated via the grade, roughness, breathiness, asthenia, and strain (GRBAS) scale. Voice-related QOL was assessed by means of the Trans Woman Voice Questionnaire (TWVQ) and the self-perceived femininity of the voice. A total of 7 patients were included. The mean age was 35.4 years, and the mean post-operative follow-up time was 13.7 months. There was a mean increase of 47.9 ± 46.6 Hz ($p = 0.023$) in sustained/e/F0 and a mean increase of 24.6 ± 27.5 Hz ($p = 0.029$) in speaking F0 following WG. There was no statistical significance in the pre- and post-operative comparison of maximum phonation time, formant frequencies, frequency range, jitter, shimmer, and grade, roughness, breathiness, asthenia, and strain scale. TWVQ decreased following surgery from 98.3 ± 9.2 to 54.1 ± 25.0 ($p = 0.007$) and mean self-perceived femininity of the voice increased from 2.8 ± 1.8 to 7.7 ± 2.4 ($p = 0.008$). One patient (14 %) presented a post-operative granuloma and there was 1 (14 %) premature suture dehiscence. The authors concluded that WG was safe and effective for feminizing the voice of transgender women. There was an increase in fundamental frequency, without aggravating other acoustic parameters or voice quality; and voice-related QOL improved after surgery.

The authors stated that this study had several limitations and weaknesses. The principal limitation was the small sample size ($n = 7$). Multi-variate regression could not reasonably be carried out in this study due to the sample size. On the other hand, the result of an increase in F0 with statistical significance -- the main outcome evaluated in the study -- associated with the improvement of subjective and objective variables (auditory-perceptual evaluation, acoustic parameters, voice-related QOL and self-perception of voice) after WG, corroborated the hypothesis that these results were not the consequence of sampling error. Furthermore, a longer follow-up period would examine if the pitch is sustained in the long-term, a possible advantage of WG over other surgical techniques, evidence that could not be ratified with a follow-up of just 6 months. These researchers believed that the most important point of this study was the prospective analysis of F0 and voice quality without post-operative voice therapy (VT), making it possible to attribute the results exclusively to WG in a more reliable manner. Furthermore, these investigators employed a specific and validated PROM for the studied population to quantify the impact of surgery on voice-related QOL.

Rapoport e al (2023) noted that currently 1-year evaluation of pitch elevation in patients undergoing modified WG in combination with VT has not been assessed. In a retrospective study, these researchers examined if 1-year pitch elevation would be sustained in patients who undergo modified WG in combination with VT for voice feminization. This trial was carried out from 2016 to 2020. Charts were reviewed for sustained vowel fundamental frequency (F0/a/), speaking fundamental frequency (SF0), and TWVQ at pre-operative, initial post-operative (3 to 6 months after surgery), and 12-month post-operative visits. Change in average F0/a/, SF0 and TWVQ was 50.25 Hz, 32.96 Hz, and 32.6 at 12-months post-operatively compared to pre-operative values. Initial and 12-month post-operative SF0 were significantly higher than pre-operative SF0 (Mann-Whitney U test $p = 0.0042$, $p = 0.0010$). There was no difference in initial post-operative and 12-month post-operative SF0 ($p = 0.50$). TWVQ at 12 months was significantly lower than pre-operative TWVQ (ANOVA $p < 0.001$, Tukey honestly significant difference HSD $p < 0.05$). The authors conclude that pitch elevation remained sustained at 1 year in patients undergoing modified WG in combination with VT. Modified WG combined with VT resulted in relatively long-term improvements in voice-related QOL and was possibly a beneficial addition in the long-term management of patients who desire voice feminization. Level of Evidence = IV.

Root et al (2023) stated that WG is a pitch-elevating surgery carried out by laryngologists to provide gender-affirming care. The surgery creates an anterior glottic web that could theoretically lead to airway concerns, either peri-operatively or at the time of future procedures; however, such concerns are not well-described in the literature. In a pilot study, these researchers examined surgeon opinions on airway concerns regarding WG. They carried out a survey of laryngologists assessing opinions on airway considerations in glottoplasty. A total of 19 physicians responded, representing approximately 193 surgeries; 52.6 % carried out glottoplasty and the remainder responded based on experience with anterior glottic webs. Two peri-operative airway complications were reported, both mild stridor that did not prevent same-day discharge. No long-term sequela was reported. All surveyed laryngologists endorsed an altered general anesthetic approach for future procedures, with 73.7 % advocating the use of a smaller endotracheal tube; 72.2 % did not have "major concerns" regarding future intubations; and only 5.3 % thought the immediate risk of airway

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compromise was a "real concern"; and 91.9 % counseled their patients routinely but briefly on airway concerns. Open-ended comments conveyed themes of concern for post-operative disruption of the web more than of airway compromise. The authors concluded that because glottoplasty is carried out in the anterior glottis and does not significantly impact airway patency, the risk of serious airway complications appears to be minimal. These investigators noted that laryngologists believe future intubations require a modified approach with a smaller tube, partly due to concern for glottic web trauma. These researchers stated that based on the findings of this pilot study, the topic deserves further investigations to standardize care and anesthetic alterations for patients with WG. Level of Evidence = V.

Rogalska et al (2024) noted that since new evidence regarding the impact of WG on the voice in transgender women became available in the literature in recent years, these investigators carried out an updated systematic review and meta-analysis to examine the safety and effectiveness of WG in the process of vocal feminization. They searched PubMed, Embase, and Cochrane for English-language studies published until July 4, 2023. Studies were found eligible if they examined the impact of WG on the acoustic-aerodynamic measures and quality of voice in transgender women. A total of 23 studies were identified. After exclusion of 3 studies due to incomplete data, 20 studies including 656 patients were included in the meta-analysis. Following WG, there was a significant increase of fundamental frequency, speaking fundamental frequency, and lower limit of the frequency range ($p < 0.001$). Concurrently, a significant reduction of frequency range and maximum phonation time was observed ($p < 0.001$). No significant differences were observed between the pre- and post-operative values regarding the GRBAS scale score ($p = 0.339$). The overall score in the TWVQ significantly improved after WG ($p < 0.001$). The authors concluded that WG was an effective voice feminization method in transgender women, associated with a high procedural success and low risk of post-operative complications. Significantly improved TWVQ score after surgery suggested its positive impact on the voice-related QOL. Post-operative decrease of maximum phonation time and frequency range did not appear to significantly impact the effectiveness of voice production.

McBrinn et al (2024) discussed the challenges faced by transgender women in achieving congruence between gender identity and voice characteristics and emphasized the importance of a multi-disciplinary approach involving therapeutic and surgical interventions. These investigators stated that recent literature on gender-affirming vocal care emphasized key themes such as assessment, voice feminization therapy, as well as various surgical techniques, including glottoplasty and its modifications. Recent publications focused on outcomes, duration, and impact on vocal quality and scrutinized complications associated with surgical interventions. Most publications in the past 18 months advocated for the integration of VT and surgery for optimal outcomes. The use of a combined approach was significantly more effective in terms of fundamental frequency gain compared to therapy alone. These researchers noted that modified WG remains the preferred and most reliable surgical intervention; however, surgery is not without its complications and vocal trade-offs, especially its impact on vocal projection and the risk of long-term dysphonia. Post-surgery rehabilitation improves long-term outcomes. Recent publications on alternative surgical approaches, namely laser reduction glottoplasty and feminization laryngoplasty, showed promise; however, the results were more difficult to generalize. The authors concluded that prospective, multi-center studies with standardized protocols are needed to establish best evidence-based practices.

Furthermore, an UpToDate review on “Gender-affirming surgery: Feminizing procedures” (Ferrando, 2025) does not mention glottoplasty as a management option.

Appendix

DSM 5 Criteria for Gender Dysphoria in Adults and Adolescents

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I. A marked incongruence between one's experienced/expressed gender and assigned gender, of at least 6 months duration, as manifested by two or more of the following:

   A. A marked incongruence between one's experienced/expressed gender and primary and/or secondary sex characteristics (or, in young adolescents, the anticipated secondary sex characteristics)
   B. A strong desire to be rid of one's primary and/or secondary sex characteristics because of a marked incongruence with one's experienced/expressed gender (or, in young adolescents, a desire to prevent the development of the anticipated secondary sex characteristics)
   C. A strong desire for the primary and/or secondary sex characteristics of the other gender
   D. A strong desire to be of the other gender (or some alternative gender different from one's assigned gender)
   E. A strong desire to be treated as the other gender (or some alternative gender different from one's assigned gender)
   F. A strong conviction that one has the typical feelings and reactions of the other gender (or some alternative gender different from one's assigned gender).

II. The condition is associated with clinically significant distress or impairment in social, occupational, or other important areas of functioning.

There is no minimum duration of relationship required with mental health professional. It is the professional's judgment as to the appropriate length of time before a referral letter can appropriately be written. A common period of time is three months, but there is significant variation in both directions.

Evaluation of candidacy for gender affirmation surgery by a mental health professional is covered under the member's medical benefit, unless the services of a mental health professional are necessary to evaluate and treat a mental health problem, in which case the mental health professional's services are covered under the member's behavioral health benefit. Please check benefit plan descriptions.

Characteristics of a Qualified Health Professionals (From SOC-8)

Qualifications of Mental Health Professional for assessing transgender and gender diverse adults for physical treatments (from WPATH SOC-8):

1. Are licensed by their statutory body and hold, at a minimum, a master's degree or equivalent training in a clinical field relevant to this role and granted by a nationally accredited statutory institution.
2. Are able to identify co-existing mental health or other psychosocial concerns and distinguish these from gender dysphoria, incongruence, and diversity.
3. Are able to assess capacity to consent for treatment.
4. Have experience or be qualified to assess clinical aspects of gender dysphoria, incongruence, and diversity.
5. Undergo continuing education in health care relating to gender dysphoria, incongruence, and diversity.
6. Liaise with professionals from different disciplines within the field of transgender health for consultation and referral on behalf of gender diverse adults seeking gender-affirming treatment, if required.

Credentials of surgeons who perform gender-affirming surgical procedures (fromWPATH SOC-8):

1. Training and documented supervision in gender-affirming procedures;
2. Maintenance of an active practice in gender-affirming surgical procedures;
3. Knowledge about gender diverse identities and expressions;
4. Continuing education in the field of gender-affirmation surgery;
5. Tracking of surgical outcomes.

Characteristics of health care professionals working with gender diverse adolescents:

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1. Are licensed by their statutory body and hold a postgraduate degree or its equivalent in a clinical field relevant to this role granted by a nationally accredited statutory institution.
2. Receive theoretical and evidenced-based training and develop expertise in general child, adolescent, and family mental health across the developmental spectrum.
3. Receive training and have expertise in gender identity development, gender diversity in children and adolescents, have the ability to assess capacity to assent/consent, and possess general knowledge of gender diversity across the life span.
4. Receive training and develop expertise in autism spectrum disorders and other neurodevelopmental presentations or collaborate with a developmental disability expert when working with autistic/neurodivergent gender diverse adolescents.
5. Continue engaging in professional development in all areas relevant to gender diverse children, adolescents, and families.

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